

Limit theorems

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The Central Limit Theorem

Remark. The sum of many independent random variables, none of which dominates the others, has a distribution that is approximately bell-shaped, whatever the distributions of the individual variables. Theorem 0.1 makes this precise.

Theorem 0.1 (Central Limit Theorem). *Let $X_1, X_2, \dots$ be IID random variables with mean μ and finite variance $\sigma^2 > 0$, and let $S_n \stackrel{\text{def}}{=} \sum_{i=1}^n X_i$. Then for every real number z :*

$$\lim_{n \rightarrow \infty} \Pr\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq z\right) = \Phi(z)$$

where $\Phi(z) \stackrel{\text{def}}{=} \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du$ is the CDF of the *standard normal distribution* $N(0, 1)$.

Remark. This version is the Lindeberg–Lévy CLT; its proof is beyond these notes' scope (Billingsley 1995, Theorem 27.1). Other versions relax the IID assumption, which is why the informal statement asks only that no summand dominate.

Corollary 0.1 (Mean and variance of a sum of IID random variables). *Let $X_1, \dots, X_n$ be IID random variables, each discrete or continuous, with mean μ and finite variance σ^2 , and let $S_n \stackrel{\text{def}}{=} \sum_{i=1}^n X_i$. Then:*

$$\begin{aligned} \mathbb{E}[S_n] &= n\mu \\ \text{Var}(S_n) &= n\sigma^2 \end{aligned}$$

i Proof

Proof. For the mean, apply [linearity of expectation](#) once per added summand:

$$\begin{aligned} \mathbb{E}[S_n] &= \sum_{i=1}^n \mathbb{E}[X_i] \quad (\text{linearity of expectation, applied } n-1 \text{ times}) \\ &= n\mu \quad (\text{each } X_i \text{ has mean } \mu) \end{aligned}$$

For the variance, apply the [variance of a linear combination](#) with every $a_i = 1$. For $i \neq j$, X_i and X_j are independent (take $A_k = \mathbb{R}$ for every other k in the [definition of independence](#)), so $\text{Cov}(X_i, X_j) = 0$ for [independent summands](#); and $\text{Cov}(X_i, X_i) = \text{Var}(X_i)$ ([covariance of a variable with itself](#)):

$$\begin{aligned} \text{Var}(S_n) &= \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) && (\text{variance of a linear combination, all } a_i = 1) \\ &= \sum_{i=1}^n \text{Cov}(X_i, X_i) + \sum_{\substack{i,j=1 \\ i \neq j}}^n \text{Cov}(X_i, X_j) && (\text{split off the terms with } i = j) \\ &= \sum_{i=1}^n \text{Var}(X_i) + 0 && (\text{covariance with itself; independent summands}) \\ &= n\sigma^2 && (\text{each } X_i \text{ has variance } \sigma^2) \end{aligned}$$

□

Remark. In practice, Theorem [0.1](#) justifies approximating S_n by a normal distribution with mean $n\mu$ and variance $n\sigma^2$, which by Corollary [0.1](#) are exactly the mean and variance of S_n .

Example 0.1 (The sum of five dice). A single fair die roll has the discrete uniform distribution on $\{1, \dots, 6\}$, which is flat, not bell-shaped (Figure [1](#)). Its mean is $\mu = 3.5$, and its variance is $\sigma^2 = \sum_{x=1}^6 (x - 3.5)^2 / 6 = 35/12$.

```

dice_sum_pmf <- function(n_dice) {
  totals <- rowSums(expand.grid(rep(list(1:6), n_dice)))
  probs <- prop.table(table(totals))
  data.frame(total = as.numeric(names(probs)), p = as.vector(probs))
}

dice_plot <-
  ggplot2::ggplot(mapping = ggplot2::aes(x = total, y = p)) +
  ggplot2::xlab("sum of dice (x)") +
  ggplot2::ylab("Probability of outcome, Pr(X=x)") +
  ggplot2::expand_limits(y = 0)

dice_plot + ggplot2::geom_col(data = dice_sum_pmf(1))

```

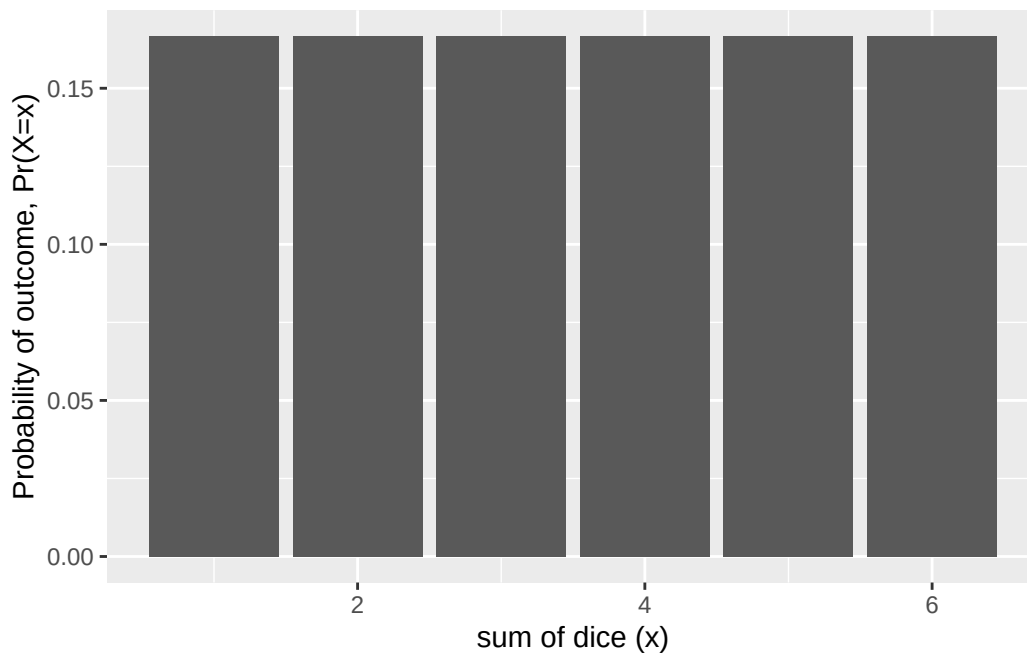


Figure 1: Distribution of the outcome of one die

The sum of five independent rolls, S_5 , is already close to bell-shaped (Figure 2).

```

pmf_five_dice <- dice_sum_pmf(5)
dice_plot + ggplot2::geom_col(data = pmf_five_dice)

```

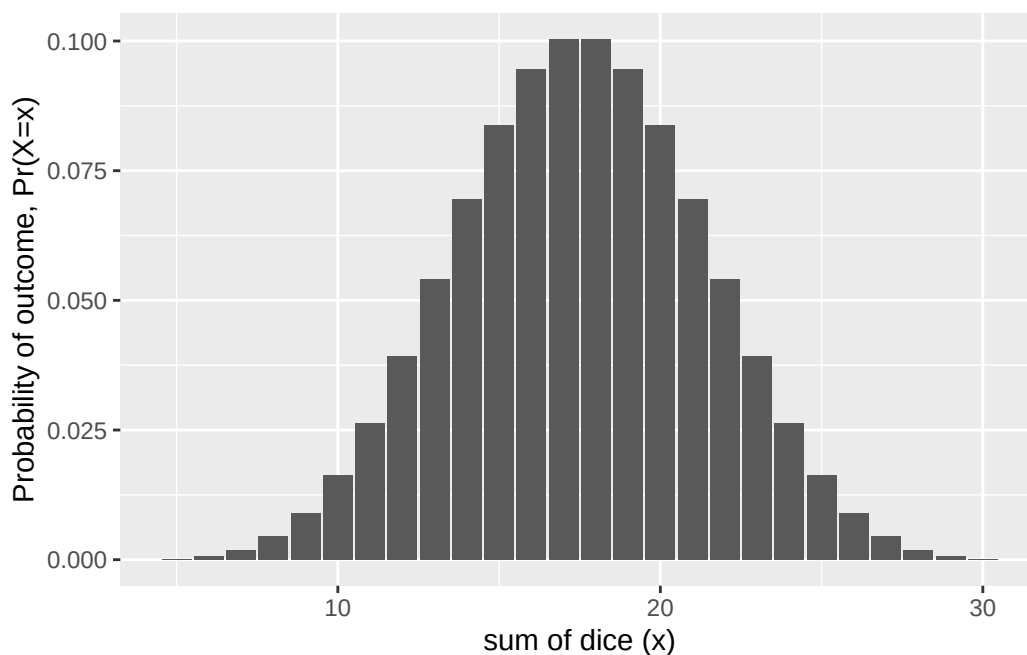


Figure 2: Distribution of the sum of five dice

For example, the exact probability that five dice total at most 15 is $\Pr(S_5 \leq 15) = 0.3052$. The normal approximation that Theorem 0.1 suggests, with mean $5 \cdot 3.5 = 17.5$ and variance $5 \cdot 35/12 \approx 14.58$, evaluated at 15.5 to account for S_5 taking only integer values, gives $\Phi((15.5 - 17.5)/\sqrt{14.58}) \approx 0.3002$.

References

Billingsley, Patrick. 1995. *Probability and Measure*. 3rd ed. Wiley Series in Probability and Mathematical Statistics. Wiley.